Construction and Properties of the Cantor Set

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May 17, 2026

Abstract

In this paper, we construct the Cantor set $C \subset [0,1]$ as the intersection of a sequence of finite unions of closed intervals, where at each step we remove the open middle third of every interval. We then prove five key properties of the Cantor set: C is compact, has measure zero, is perfect, is totally disconnected, and is uncountable. The Cantor set is an important example in mathematics which challenges intuition about relationships between concepts in cardinality, topology, and measure theory.

1 Introduction

In this paper, we construct the Cantor set C and prove the five key properties stated above. First, we define an operator T which removes the open middle third of each interval within a finite disjoint union of closed intervals. Then, we define the Cantor sequence (C_n) by setting $C_1 = [0,1]$ and applying T iteratively. Finally, we define the Cantor set as the intersection of this sequence. The Cantor set is important because it holds several seemingly contradictory properties at once; the set has no length, yet uncountably many points and it has no isolated points, yet is totally disconnected. By pushing the boundaries of these concepts, we can better understand their relationships and scopes.

The paper is organized as follows. Section 2 constructs C and proves a lemma describing the intermediate sets C_n. Section 3 proves the five key properties. This exposition follows the treatment in [1].

2 Constructing the Cantor set

My construction follows the same middle-thirds idea as [1], but is organized around the sequence of remaining closed sets (C_n) rather than the removed open sets.

Definition 2.1. For a closed interval $[a,b] \subset \mathbb{R}$ with $a \leq b$, define

$$T([a,b]) := \left[a, a + \tfrac{b-a}{3}\right] \cup \left[b - \tfrac{b-a}{3}, b\right].$$

If $S = \bigcup_{i=1}^{n} I_i$ is a finite disjoint union of closed intervals, extend T componentwise by

$$T(S) := \bigcup_{i=1}^{n} T(I_i).$$

Definition 2.2. Set $C_1 = [0, 1]$ and $C_{n+1} = T(C_n)$ for $n \geq 1$. We will refer to this as the Cantor sequence. The Cantor set is

$$C := \bigcap_{n=0}^{\infty} C_n.$$

Example 2.3. In the example, we can see the first eight iterations of the Cantor sequence. In the image, the black bars represent a closed interval. Notice that the number of intervals increases and the intervals become shorter as the iteration count increases. Also notice the recursive pattern; if we were to zoom in on $C_n \cap [0, \frac{1}{3}]$, the image would reveal a scaled copy of C_n.

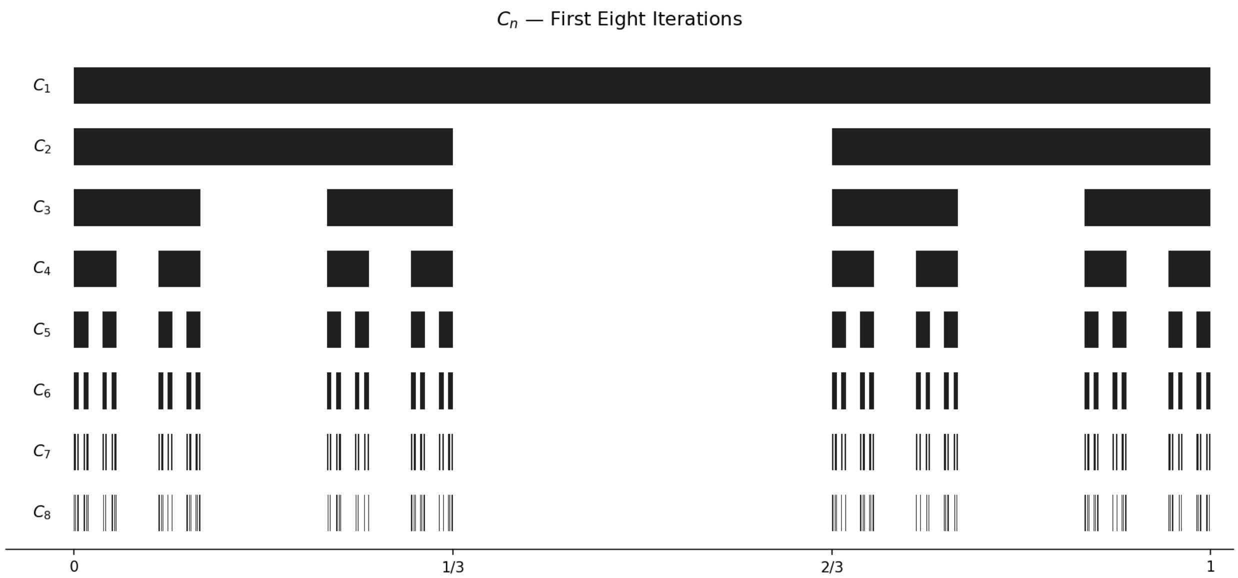

Figure 1: The first eight iterations of the Cantor sequence.

Lemma 2.4. *For each $n \geq 1$, the set C_n is a disjoint union of 2^n closed intervals, each of length 3^{-n}. Moreover, $C_{n+1} \subset C_n$.*

Proof. □

3 Properties of the Cantor set

Theorem 3.1. *C is compact.*

Proof. □

Lemma 3.2. *C is nonempty.*

Proof. □

Definition 3.3 (Measure Zero)**.** A set S is measure zero if for every $\epsilon > 0$, S can be covered by finitely many intervals of total length less than ϵ.

Theorem 3.4. *C has measure zero.*

Proof. ∎

Definition 3.5 (Isolated point). Let (M, d) be a metric space and $S \subset M$. $x \in S$ is an isolated point of S if there exists an $\epsilon > 0$ such that $B_\epsilon(x) \cap S = \{x\}$. This can be interpreted as a resolution (ϵ) existing such that x has no nearby points in S.

Definition 3.6 (Perfect). A set S is perfect if it is closed and has no isolated points.

Theorem 3.7. *C is perfect.*

Proof. ∎

Theorem 3.8. *C is totally disconnected (its only connected subsets are singletons).*

Proof. ∎

Theorem 3.9. *C is uncountable*

Proof. ∎

4 Conclusion

References

[1] R. Johnsonbaugh and W. E. Pfaffenberger, *Foundations of Mathematical Analysis*, Dover Publications, 2002.